

Chapter 3 — Motion in Central Potential $V(r)$; the Kepler Problem

HC 6001 – Dr. Castillo

15 October 2024

Chapter 3 — Motion in central potential $V(r)$

$$\begin{aligned}
 \mathbf{L} &= \mu r^2 \dot{\phi} \hat{\mathbf{z}} & \mathbf{r} &= r \hat{\mathbf{r}} & \mathbf{p} &= \mu \dot{\mathbf{r}} \\
 \mathbf{L} &= \text{const. of motion} \\
 L &= T - V = \frac{\mu}{2} (\dot{r}^2 + r^2 \dot{\phi}^2) - V(r) \\
 \phi \text{ cyclic} &\iff \frac{\partial L}{\partial \phi} = 0 \implies p_\phi = \frac{\partial L}{\partial \dot{\phi}} = \mu r^2 \dot{\phi} \equiv \ell && \text{CONSERVED} \\
 V_{\text{EFF}}(r) &= V(r) + \frac{\ell^2}{2\mu r^2} \\
 E = T + V &= \frac{\mu}{2} (\dot{r}^2 + r^2 \dot{\phi}^2) + V(r) = \frac{\mu}{2} \dot{r}^2 + V_{\text{EFF}}(r)
 \end{aligned}$$

Universal method $\Rightarrow$ complete solution

$$\begin{array}{ll}
 t = \int_{r_0}^r \frac{dr'}{\sqrt{\frac{2}{\mu} \left(E - V(r') - \frac{\ell^2}{2\mu r'^2} \right)}} & t = t(r) \\
 \phi = \phi_0 + \int_r^t \frac{\ell}{\mu r^2(t')} dt' & \phi = \phi(t)
 \end{array}$$

Remark 1 (on the second quadrature). The lower limit of the t' -integral is written on the board with the lecturer's Γ -shaped r , not with his crossed t ; it is reproduced here as written. The differential is cramped and reads at first glance as “ At' ”: it is dt' .

Left-hand side of the radial equation, in the variable u :

$$\mu \ddot{r} = -\frac{\ell^2}{\mu} \frac{1}{r^2} \frac{d}{d\phi} \frac{du}{d\phi} = -\frac{\ell^2}{\mu} u^2 \frac{d^2 u}{d\phi^2}$$

Diff. eqn. for $r = r(\phi)$

$$\begin{aligned}
 \ell = \mu r^2 \dot{\phi} &\implies \dot{\phi} = \frac{\ell}{\mu r^2} = \frac{d\phi}{dt} \\
 \frac{dr}{dt} &= \frac{dr}{d\phi} \frac{d\phi}{dt} = \frac{\ell}{\mu r^2} \frac{dr}{d\phi}
 \end{aligned}$$

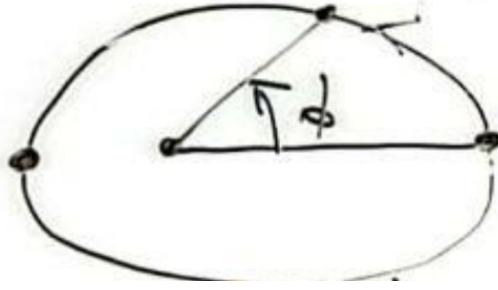


Figure 1: The orbit and the polar angle ϕ measured from the focus.

Radial eq. of motion

$$\begin{aligned}\mu \ddot{r} &= -\frac{\partial V_{\text{EFF}}}{\partial r} = -\frac{\partial}{\partial r} \left(V(r) + \frac{\ell^2}{2\mu r^2} \right) \\ \mu \ddot{r} &= F(r) + \frac{\ell^2}{\mu r^3} \quad \frac{\partial}{\partial r}(r^{-2}) = (-2)r^{-3} \\ u &= \frac{1}{r}, \quad \frac{du}{d\phi} = \frac{du}{dr} \frac{dr}{d\phi} = \frac{-1}{r^2} \frac{dr}{d\phi} \\ \frac{\ell^2}{\mu r^3} &= \frac{\ell^2}{\mu} u^3 \implies \text{RHS: } F\left(\frac{1}{u}\right) + \frac{\ell^2}{\mu} u^3 \\ \text{LHS: } \mu \frac{d}{dt} \frac{dr}{dt} &= \mu \frac{d}{dt} \frac{\ell}{\mu r^2} \frac{dr}{d\phi} = \cancel{\mu} \frac{\ell}{\cancel{\mu} r^2} \frac{d}{d\phi} \left(\frac{\ell}{\mu r^2} \frac{dr}{d\phi} \right)\end{aligned}$$

LHS = RHS:

$-\frac{\ell^2}{\mu} u^2 \frac{d^2 u}{d\phi^2} = F\left(\frac{1}{u}\right) + \frac{\ell^2}{\mu} u^3$	GENERIC EQ. OF TRAJECTORY FOR ARBITRARY $V(r)$
--	--

Remark 2 (exponents). The two powers of u in the boxed line differ: u^2 multiplies $d^2u/d\phi^2$ on the left, u^3 stands alone on the right. The two glyphs are nearly identical at board scale; the left-hand power is fixed independently by the chain $\mu \ddot{r} = -(\ell^2/\mu)u^2 d^2u/d\phi^2$ recorded above.

Eq. for trajectory $r(\phi)$

$$\begin{aligned}F(r) &= -\frac{\partial V}{\partial r} \\ \frac{dV}{dr} &= \frac{dV}{du} \frac{du}{dr} = -u^2 \frac{dV}{du}, \quad \frac{du}{dr} = -\frac{1}{r^2} = -u^2 \\ F\left(\frac{1}{u}\right) &= F(r) = -\frac{dV}{dr} = u^2 \frac{dV(1/u)}{du}\end{aligned}$$

The next line is struck out on the board — a superseded intermediate step, kept here as retracted:

$$-F(1/u) = -u^2 dV(1/u)/du \quad [\text{struck out in the original}]$$

$$-F\left(\frac{1}{u}\right) = \frac{\ell^2}{\mu} u^2 \left[u + \frac{d^2 u}{d\phi^2} \right]$$

$$-\cancel{\mu} \frac{dV}{du} = \frac{\ell^2}{\mu} \cancel{\mu} \left[u + \frac{d^2 u}{d\phi^2} \right]$$

The two struck factors u^2 cancel, leaving

$\frac{d^2 u}{d\phi^2} + u = -\frac{\mu}{\ell^2} \frac{d}{du} V\left(\frac{1}{u}\right)$	EQ. FOR TRAJECTORY $r(\phi)$ solve for given V
--	---

$$u = u_0 \quad \text{for} \quad \phi = \phi_0, \quad \frac{du}{d\phi} = 0$$

$$\frac{du}{d\phi} = 0 \iff \text{max. or min. } r \text{ point}$$

Remark 3 (a slip in the second derivative). In the line $-F(1/u) = (\ell^2/\mu)u^2[u + d^2u/d\phi^2]$ the superscript 2 on the numerator d is written detached and set high above the fraction, so that the term can be misread as $du/d\phi^2$. It is written correctly and tightly on the line immediately below; both are read here as $d^2u/d\phi^2$.

Gravitation case — Kepler problem

Remark 4 (ink). GRAVITATION CASE is written in green on the board; KEPLER PROBLEM, underlined, is in blue.

$$V = -\frac{k}{r} = -ku, \quad \frac{dV}{du} = -k$$

$\frac{d^2 u}{d\phi^2} + u = \frac{k\mu}{\ell^2}$	harmonic oscillator with shifted origin
---	--

Aside.

$$\tilde{u} = u - \frac{k\mu}{\ell^2} \implies \boxed{\frac{d^2 \tilde{u}}{d\phi^2} + \tilde{u} = 0}$$

*Homogeneous solution*¹

$$\begin{aligned} \tilde{u}(\phi) &= A \cos(\phi - \phi_0), \\ \tilde{u}'(\phi) &= -A \sin(\phi - \phi_0), \\ \tilde{u}''(\phi) &= -A \cos(\phi - \phi_0) = -\tilde{u}(\phi). \end{aligned}$$

$u(\phi) = \frac{k\mu}{\ell^2} + A \cos(\phi - \phi_0)$

$u(\phi) = \frac{k\mu}{\ell^2} \left[e \cos(\phi - \phi_0) + 1 \right] = \frac{1}{r}$
--

$$e \text{ constant}, \quad e \equiv \frac{A}{k\mu/\ell^2}$$

¹The word SOLUTION runs into the board frame in the photograph and its last letters are obscured on the board itself.

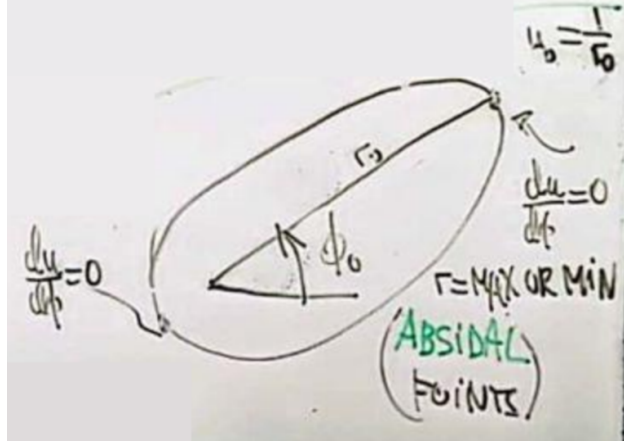


Figure 2: Apsidal points of the orbit: $u_0 = 1/r_0$, and $du/d\phi = 0$ at both ends of the major axis, where r is a maximum or a minimum. The board's spelling ABSIDAL is reproduced verbatim [editorial note: the standard spelling is *apsidal*]; the parenthesis (ABSIDAL POINTS) is written in green.

Using the universal method

$$u'' = -u + \frac{k\mu}{\ell^2} \quad \Downarrow \times u'(\phi)$$

$$\frac{du}{d\phi} = (u')^2 = \sqrt{\quad}$$

Caveat 1. The radical on this line is drawn in full — stroke and vinculum — but its interior was left empty on the board: it contains only a stray dot and a short dash, no expression. Nothing is supplied here. Note also that, as written, $du/d\phi = (u')^2$ is loose notation, since u' is $du/d\phi$; the object intended is the first integral obtained by multiplying through by $u'(\phi)$. Transcribed as written.

$$d\phi = \frac{du}{u'}$$

⋮

A vertical dotted ellipsis of about six dots runs from the line above down to the boxed result; the elided steps are not reconstructed here.

$$u = \frac{\mu k}{\ell^2} \left[1 + \sqrt{1 + \frac{2\ell^2 E}{\mu k^2}} \cos(\phi - \phi_0) \right] = \frac{1}{r}$$

$$e \equiv \sqrt{1 + \frac{2\ell^2 E}{\mu k^2}}$$

Remark 5 (scope of the radical). The vinculum terminates before $\cos(\phi - \phi_0)$: the cosine stands outside the square root, consistent with the boxed definition of e on the line below.